

δ -Bi-Cluster

□ For a submatrix $I \times J$

□ The mean of the i -th row: $e_{iJ} = \frac{1}{|J|} \sum_{j \in J} e_{ij}$

□ The mean of the j -th column: $e_{IJ} = \frac{1}{|I|} \sum_{i \in I} e_{ij}$

□ The mean of all elements in the submatrix:

$$e_{IJ} = \frac{1}{|I||J|} \sum_{i \in I, j \in J} e_{ij} = \frac{1}{|I|} \sum_{i \in I} e_{iJ} = \frac{1}{|J|} \sum_{j \in J} e_{IJ}$$

□ The quality of the submatrix as a bi-cluster can be measured by the mean squared residue value

$$H(I \times J) = \frac{1}{|I||J|} \sum_{i \in I, j \in J} (e_{ij} - e_{iJ} - e_{IJ} + e_{IJ})^2$$

δ -Bi-Cluster

- A submatrix $I \times J$ is **δ -bi-cluster** if $H(I \times J) \leq \delta$ where $\delta \geq 0$ is a threshold. When $\delta = 0$, $I \times J$ is a perfect bi-cluster with coherent values. By setting $\delta > 0$, a user can specify the tolerance of average noise per element against a perfect bi-cluster
- $\text{residue}(e_{ij}) = e_{ij} - e_{iJ} - e_{Ij} + e_{IJ}$

The δ -Cluster Algorithm

- ❑ Maximal δ -bi-cluster is a δ -bi-cluster $I \times J$ such that there does not exist another δ -bi-cluster $I' \times J'$ which contains $I \times J$
- ❑ Computing is costly: Use heuristic greedy search to obtain local optimal clusters
- ❑ Two phase computation: deletion phase and additional phase
- ❑ Deletion phase: Start from the whole matrix, iteratively remove rows and columns while the mean squared residue of the matrix is over δ
 - ❑ At each iteration, for each row/column, compute the mean squared residue:

$$d(i) = \frac{1}{|J|} \sum_{j \in J} (e_{ij} - e_{iJ} - e_{IJ} + e_{IJ})^2$$
$$d(j) = \frac{1}{|I|} \sum_{i \in I} (e_{ij} - e_{iJ} - e_{IJ} + e_{IJ})^2$$

- ❑ Remove the row or column of the largest mean squared residue

The δ -Cluster Algorithm

- ❑ Addition phase:
 - ❑ Expand iteratively the δ -bi-cluster $I \times J$ obtained in the deletion phase as long as the δ -bi-cluster requirement is maintained
 - ❑ Consider all the rows/columns not involved in the current bi-cluster $I \times J$ by calculating their mean squared residues
 - ❑ A row/column of the smallest mean squared residue is added into the current δ -bi-cluster
- ❑ It finds only one δ -bi-cluster, thus needs to run multiple times: replacing the elements in the output bi-cluster by random numbers